**BILLING ALERT**

Simple Notification Service

**Subscription confirmed!**

You have successfully subscribed.

Your subscription's id is:
arn:aws:sns:us-east-1:768499314195:BillingAlertTopic:fa573216-758f-42c8-ab49-c86b5f68a6f1

If it was not your intention to subscribe, click here to unsubscribe.

## Installation

us-east-1 +

```
~ $ aws sts get-caller-identity
{
    "UserId": "768499314195",
    "Account": "768499314195",
    "Arn": "arn:aws:iam::768499314195:root"
}
~ $
```

**GITHUB CLONED**

```
~ $ git clone https://github.com/icdfa/GRC208-AWS-Capstone-Project.git
Cloning into 'GRC208-AWS-Capstone-Project'...
remote: Enumerating objects: 60, done.
remote: Counting objects: 100% (60/60), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 60 (delta 16), reused 56 (delta 12), pack-reused 0 (from 0)
Receiving objects: 100% (60/60), 2.52 MiB | 43.05 MiB/s, done.
Resolving deltas: 100% (16/16), done.
~ $ cd GRC208-AWS-Capstone-Project
GRC208-AWS-Capstone-Project $ ls -la
total 276
drwxr-xr-x. 4 cloudshell-user cloudshell-user  4096 Mar 31 16:40 .
drwxrwxrwx. 5 cloudshell-user cloudshell-user  4096 Mar 31 16:40 ..
-rw-r--r--. 1 cloudshell-user cloudshell-user  3551 Mar 31 16:40 architecture_design.md
-rw-r--r--. 1 cloudshell-user cloudshell-user  2183 Mar 31 16:40 architecture-diagram.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 13187 Mar 31 16:40 AWS_SERVICES_GUIDE.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 11185 Mar 31 16:40 BEST_PRACTICES.md
-rw-r--r--. 1 cloudshell-user cloudshell-user  8810 Mar 31 16:40 cloudformation-database-stack.yaml
-rw-r--r--. 1 cloudshell-user cloudshell-user  6959 Mar 31 16:40 cloudformation-network-stack.yaml
-rw-r--r--. 1 cloudshell-user cloudshell-user 11238 Mar 31 16:40 DELIVERY_SUMMARY.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 11945 Mar 31 16:40 DEPLOYMENT_GUIDE.md
-rwxr-xr-x. 1 cloudshell-user cloudshell-user  6985 Mar 31 16:40 deploy.sh
drwxr-xr-x. 2 cloudshell-user cloudshell-user  4096 Mar 31 16:40 diagrams
drwxr-xr-x. 7 cloudshell-user cloudshell-user  4096 Mar 31 16:40 .git
-rw-r--r--. 1 cloudshell-user cloudshell-user   545 Mar 31 16:40 .gitignore
-rw-r--r--. 1 cloudshell-user cloudshell-user 10995 Mar 31 16:40 grc-dashboard.css
-rw-r--r--. 1 cloudshell-user cloudshell-user 11979 Mar 31 16:40 grc-dashboard.jsx
-rw-r--r--. 1 cloudshell-user cloudshell-user  9996 Mar 31 16:40 lambda_compliance_monitor.py
-rw-r--r--. 1 cloudshell-user cloudshell-user 15630 Mar 31 16:40 PRESENTATION_GUIDE.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 48935 Mar 31 16:40 PRESENTATION_SCRIPT.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 10971 Mar 31 16:40 PROJECT_MANIFEST.md
-rw-r--r--. 1 cloudshell-user cloudshell-user 21421 Mar 31 16:40 README.md
-rw-r--r--. 1 cloudshell-user cloudshell-user   175 Mar 31 16:40 requirements.txt
-rw-r--r--. 1 cloudshell-user cloudshell-user 10489 Mar 31 16:40 sample_data.sql
-rw-r--r--. 1 cloudshell-user cloudshell-user 13656 Mar 31 16:40 test_cases.py
GRC208-AWS-Capstone-Project $ ~ aws cloudformation deploy \
```

# Phase 1: Network Infrastructure

```
/home/cloudshell-user/GRC208-AWS-Capstone-Project
GRC208-AWS-Capstone-Project $ cd ~/GRC208-AWS-Capstone-Project
GRC208-AWS-Capstone-Project $ aws cloudformation deploy --stack-name grc-capstone-network-stack --template-file cloudformation-network-stack.yaml --capabilities CAPABILITY_NAMED_IAM --parameter-overrides EnvironmentName
=grc-capstone --region us-east-1

Waiting for changeset to be created..
Waiting for stack create/update to complete
Successfully created/updated stack - grc-capstone-network-stack
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws cloudformation deploy --stack-name g
=grc-capstone --region us-east-1

Waiting for changeset to be created..
Waiting for stack create/update to complete
Successfully created/updated stack - grc-capstone-network-stack
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws cloudformation describe-stacks --stack-name grc-capstone-network-stack --query "Stacks[0].StackStatus" --output text
CREATE_COMPLETE
GRC208-AWS-Capstone-Project $
```

# Phase 2: Database Infrastructure

## S3

```
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-cloudtrail-logs-768499314195 --region us-east-1
{
    "Location": "/grc-cloudtrail-logs-768499314195",
    "BucketArn": "arn:aws:s3:::grc-cloudtrail-logs-768499314195"
}
GRC208-AWS-Capstone-Project $ aws s3 ls | grep grc-capstone
2026-03-31 19:35:09 grc-capstone-compliance-reports-768499314195
2026-03-31 19:35:08 grc-capstone-evidence-bucket-768499314195
GRC208-AWS-Capstone-Project $ aws s3 ls | grep grc-
2026-03-31 19:35:09 grc-capstone-compliance-reports-768499314195
2026-03-31 19:35:08 grc-capstone-evidence-bucket-768499314195
2026-03-31 19:35:11 grc-cloudtrail-logs-768499314195
2026-03-31 19:35:10 grc-config-bucket-768499314195
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws cloudformation delete-stack --stack-name grc-capstone-database-stack --region us-east-1
GRC208-AWS-Capstone-Project $ # Evidence bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-capstone-evidence-bucket-768499314195 --region us-east-1

{
    "Location": "/grc-capstone-evidence-bucket-768499314195",
    "BucketArn": "arn:aws:s3:::grc-capstone-evidence-bucket-768499314195"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Reports bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-capstone-compliance-reports-768499314195 --region us-east-1
{
    "Location": "/grc-capstone-compliance-reports-768499314195",
    "BucketArn": "arn:aws:s3:::grc-capstone-compliance-reports-768499314195"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Config bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-config-bucket-768499314195 --region us-east-1
{
    "Location": "/grc-config-bucket-768499314195",
    "BucketArn": "arn:aws:s3:::grc-config-bucket-768499314195"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # CloudTrail logs bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-cloudtrail-logs-768499314195 --region us-east-1
{
    "Location": "/grc-cloudtrail-logs-768499314195",
    "BucketArn": "arn:aws:s3:::grc-cloudtrail-logs-768499314195"
}
GRC208-AWS-Capstone-Project $
```


```
GRC208-AWS-Capstone-Project $ # Config bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-config-bucket-768499314195 --region us-east-1
{
    "Location": "/grc-config-bucket-768499314195",
    "BucketArn": "arn:aws:s3:::grc-config-bucket-768499314195"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # CloudTrail logs bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-cloudtrail-logs-768499314195 --region us-east-1
{
    "Location": "/grc-cloudtrail-logs-768499314195",
    "BucketArn": "arn:aws:s3:::grc-cloudtrail-logs-768499314195"
}
GRC208-AWS-Capstone-Project $
```

## CLOUDTRAIL

```
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # CloudTrail logs bucket
GRC208-AWS-Capstone-Project $ aws s3api create-bucket --bucket grc-cloudtrail-logs-768499314195 --region us-east-1
{
    "Location": "/grc-cloudtrail-logs-768499314195",
    "BucketArn": "arn:aws:s3:::grc-cloudtrail-logs-768499314195"
}
GRC208-AWS-Capstone-Project $ # Controls table
GRC208-AWS-Capstone-Project $ aws dynamodb create-table \
>   --table-name grc-capstone-controls-768499314195 \
>   --attribute-definitions AttributeName=control_id,AttributeType=S \
>   --key-schema AttributeName=control_id,KeyType=HASH \
>   --billing-mode PAY_PER_REQUEST \
>   --region us-east-1

{
    "TableDescription": {
        "AttributeDefinitions": [
            {
                "AttributeName": "control_id",
                "AttributeType": "S"
            }
        ],
        "TableName": "grc-capstone-controls-768499314195",
        "KeySchema": [
            {
                "AttributeName": "control_id",
                "KeyType": "HASH"
            }
        ],
        "TableStatus": "CREATING",
        "CreationDateTime": "2026-03-31T19:36:23.457000+00:00",
        "ProvisionedThroughput": {
            "NumberOfDecreasesToday": 0,
            "ReadCapacityUnits": 0,
            "WriteCapacityUnits": 0
        }
```

```
        },
        "TableSizeBytes": 0,
        "ItemCount": 0,
        "TableArn": "arn:aws:dynamodb:us-east-1:768499314195:table/grc-capstone-controls-768499314195",
        "TableId": "e5c10fb3-7b35-4e02-bfa9-0621df378507",
        "BillingModeSummary": {
            "BillingMode": "PAY_PER_REQUEST"
        },
        "DeletionProtectionEnabled": false
    }
}
```

## AWS DYNAMODB TABLE

```
GRC208-AWS-Capstone-Project $ # Risk table
GRC208-AWS-Capstone-Project $ aws dynamodb create-table \
>   --table-name grc-capstone-risk-768499314195 \
>   --attribute-definitions AttributeName=risk_id,AttributeType=S \
>   --key-schema AttributeName=risk_id,KeyType=HASH \
>   --billing-mode PAY_PER_REQUEST \
>   --region us-east-1
{
    "TableDescription": {
        "AttributeDefinitions": [
            {
                "AttributeName": "risk_id",
                "AttributeType": "S"
            }
        ],
        "TableName": "grc-capstone-risk-768499314195",
        "KeySchema": [
            {
                "AttributeName": "risk_id",
                "KeyType": "HASH"
            }
        ],
        "TableStatus": "CREATING",
        "CreationDateTime": "2026-03-31T19:36:24.537000+00:00",
        "ProvisionedThroughput": {
            "NumberOfDecreasesToday": 0,
            "ReadCapacityUnits": 0,
            "WriteCapacityUnits": 0
        },
        "TableSizeBytes": 0,
        "ItemCount": 0,
        "TableArn": "arn:aws:dynamodb:us-east-1:768499314195:table/grc-capstone-risk-768499314195",
        "TableId": "994b18f0-0bbd-457f-97fb-0a5208194eaa",
        "BillingModeSummary": {
            "BillingMode": "PAY_PER_REQUEST"
        },
        "DeletionProtectionEnabled": false
    }
```

## AWS DYNAMODB COMPLIANCE TABLE

```
GRC208-AWS-Capstone-Project $ aws dynamodb create-table \
>   --table-name grc-capstone-compliance-768499314195 \
>   --attribute-definitions AttributeName=framework_id,AttributeType=S \
>   --key-schema AttributeName=framework_id,KeyType=HASH \
>   --billing-mode PAY_PER_REQUEST \
>   --region us-east-1
{
    "TableDescription": {
        "AttributeDefinitions": [
            {
                "AttributeName": "framework_id",
                "AttributeType": "S"
            }
        ],
        "TableName": "grc-capstone-compliance-768499314195",
        "KeySchema": [
            {
                "AttributeName": "framework_id",
                "KeyType": "HASH"
            }
        ],
        "TableStatus": "CREATING",
        "CreationDateTime": "2026-03-31T19:36:25.638000+00:00",
        "ProvisionedThroughput": {
            "NumberOfDecreasesToday": 0,
            "ReadCapacityUnits": 0,
            "WriteCapacityUnits": 0
        },
        "TableSizeBytes": 0,
        "ItemCount": 0,
        "TableArn": "arn:aws:dynamodb:us-east-1:768499314195:table/grc-capstone-compliance-768499314195",
        "TableId": "f9b775a9-700d-47e7-8484-f6656f69b887",
        "BillingModeSummary": {
            "BillingMode": "PAY_PER_REQUEST"
        },
        "DeletionProtectionEnabled": false
    }
}
```

## RDS CREATION

```
GRC208-AWS-Capstone-Project $ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].DBInstanceStatus" --output text --region us-east-1
available
GRC208-AWS-Capstone-Project $ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].DBInstanceStatus" --output text --region us-east-1
available
GRC208-AWS-Capstone-Project $ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].{DBInstanceIdentifier:DBInstanceIdentifier,DBInstanceStatus:DBInstanceStatus,DBInstanceClass:D
BInstanceClass,Engine:Engine,EngineVersion:EngineVersion,MasterUsername:MasterUsername,AllocatedStorage:AllocatedStorage,BackupRetentionPeriod:BackupRetentionPeriod,PubliclyAccessible:PubliclyAccessible,Endpoint:Endpoin
t.Address}" --output table --region us-east-1
-----------------------------------------------------------------------------
|                          DescribeDBInstances                              |
+-----------------------+---------------------------------------------------+
|  AllocatedStorage     |  20                                               |
|  BackupRetentionPeriod |  0                                               |
|  DBInstanceClass      |  db.t3.micro                                      |
|  DBInstanceIdentifier |  grc-capstone-db                                  |
|  DBInstanceStatus     |  available                                        |
|  Endpoint             |  grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com |
|  Engine               |  mysql                                            |
|  EngineVersion        |  8.0.44                                           |
|  MasterUsername       |  grcadmin                                         |
|  PubliclyAccessible   |  False                                            |
+-----------------------+---------------------------------------------------+
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].{Status:DBInstanceStatus,Endpoint:Endpoint.Address,ARN:DBInstanceArn}" --output table --regio
 us-east-1
-----------------------------------------------------------------------------------------------------------------
|                                                DescribeDBInstances                                              |
+-------------------------------------------------------+-------------------------------------------------------+------------+
|                          ARN                          |                       Endpoint                        |  Status    |
+-------------------------------------------------------+-------------------------------------------------------+------------+
|  arn:aws:rds:us-east-1:768499314195:db:grc-capstone-db |  grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com |  available |
+-------------------------------------------------------+-------------------------------------------------------+------------+
GRC208-AWS-Capstone-Project $
```

```
-----------------------------------------------------------------------------------------------------------------
|                                                DescribeDBInstances                                              |
+-------------------------------------------------------+-------------------------------------------------------+------------+
|                          ARN                          |                       Endpoint                        |  Status    |
+-------------------------------------------------------+-------------------------------------------------------+------------+
|  arn:aws:rds:us-east-1:768499314195:db:grc-capstone-db |  grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com |  available |
+-------------------------------------------------------+-------------------------------------------------------+------------+
GRC208-AWS-Capstone-Project $
```

## RESOURCE SUMMARY

```
=== GRC208 CAPSTONE - COMPLETE RESOURCE SUMMARY ===

AWS Account: 768499314195
Region: us-east-1

1. RDS Database:
-----------------------------------------------------------------------------
|                            DescribeDBInstances                            |
+-------------------------------------------------------------+------------+
|                           Endpoint                          |  Status    |
+-------------------------------------------------------------+------------+
|  grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com   |  available |
+-------------------------------------------------------------+------------+

2. S3 Buckets:
2026-03-31 19:35:09 grc-capstone-compliance-reports-768499314195
2026-03-31 19:35:08 grc-capstone-evidence-bucket-768499314195

3. DynamoDB Tables:
----------------------------------------
|               ListTables               |
+----------------------------------------+
|  grc-capstone-compliance-768499314195  |
|  grc-capstone-controls-768499314195    |
|  grc-capstone-risk-768499314195        |
+----------------------------------------+

4. CloudWatch Alarms:
grc-capstone-billing-alert-10

5. CloudTrail:

6. EventBridge Rule:
```

## LAMBDA

```
# Create IAM role for Lambda
```

```
GRC208-AWS-Capstone-Project $ aws iam create-role \
>   --role-name grc-lambda-execution-role \
>   --assume-role-policy-document file://lambda-trust-policy.json \
>   --region us-east-1
{
    "Role": {
        "Path": "/",
        "RoleName": "grc-lambda-execution-role",
        "RoleId": "AROA3F3RA5YJUQ6QBSASD",
        "Arn": "arn:aws:iam::768499314195:role/grc-lambda-execution-role",
        "CreateDate": "2026-03-31T20:09:53+00:00",
        "AssumeRolePolicyDocument": {
            "Version": "2012-10-17",
            "Statement": [
                {
                    "Effect": "Allow",
                    "Principal": {
                        "Service": "lambda.amazonaws.com"
                    },
                    "Action": "sts:AssumeRole"
                }
            ]
        }
    }
}
GRC208-AWS-Capstone-Project $
```

```
# Attach policies to role
```

```
GRC208-AWS-Capstone-Project $ aws iam put-role-policy \
>   --role-name grc-lambda-execution-role \
>   --policy-name grc-lambda-permissions \
>   --policy-document file://lambda-permissions-updated.json \
>   --region us-east-1
GRC208-AWS-Capstone-Project $ # Test Lambda function
GRC208-AWS-Capstone-Project $ aws lambda invoke \
>   --function-name grc-compliance-monitor \
>   --region us-east-1 \
>   response.json

{
    "StatusCode": 200,
    "ExecutedVersion": "$LATEST"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check the response
GRC208-AWS-Capstone-Project $ cat response.json
{"statusCode": 200, "body": "{\"message\": \"Compliance monitoring completed\", \"compliance_percentage\": 0, \"non_compliant_rules\": 0}"}GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws iam put-role-policy \
>   --role-name grc-lambda-execution-role \
>   --policy-name grc-lambda-permissions \
>   --policy-document file://lambda-permissions.json \
>   --region us-east-1
GRC208-AWS-Capstone-Project $ sleep 10
GRC208-AWS-Capstone-Project $ # First, create the deployment package if not already done
GRC208-AWS-Capstone-Project $ mkdir -p lambda-package && cp lambda_compliance_monitor.py lambda-package/ && cd lambda-package && zip -r ../lambda-function.zip . && cd ..
updating: lambda_compliance_monitor.py (deflated 74%)
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Create Lambda function with the new role
GRC208-AWS-Capstone-Project $ aws lambda create-function \
>   --function-name grc-compliance-monitor \
>   --runtime python3.11 \
>   --role arn:aws:iam::768499314195:role/grc-lambda-execution-role \
>   --handler lambda_compliance_monitor.lambda_handler \
>   --zip-file fileb://lambda-function.zip \
>   --timeout 60 \
>   --memory-size 256 \
>   --region us-east-1
{
    "FunctionName": "grc-compliance-monitor",
    "FunctionArn": "arn:aws:lambda:us-east-1:768499314195:function:grc-compliance-monitor",
    "Runtime": "python3.11",
    "Role": "arn:aws:iam::768499314195:role/grc-lambda-execution-role",
    "Handler": "lambda_compliance_monitor.lambda_handler",
    "CodeSize": 2765,
    "Description": "",
    "Timeout": 60,
    "MemorySize": 256,
    "LastModified": "2026-03-31T20:12:10.196+0000",
    "CodeSha256": "FAn4JoNwg5WAUkrt64klEPizDq3iXvZOAYmrn/k6578=",
    "Version": "$LATEST",
    "TracingConfig": {
        "Mode": "PassThrough"
    },
    "RevisionId": "c63e0a93-36c1-4896-9fb9-ce35659964c9",
    "State": "Pending",
```

```
    "StateReason": "The function is being created.",
    "StateReasonCode": "Creating",
    "PackageType": "Zip",
    "Architectures": [
        "x86_64"
    ],
    "EphemeralStorage": {
        "Size": 512
    },
    "SnapStart": {
        "ApplyOn": "None",
        "OptimizationStatus": "Off"
    },
    "RuntimeVersionConfig": {
        "RuntimeVersionArn": "arn:aws:lambda:us-east-1::runtime:f14046b558b69f092cf3c888cd59ed30350756eb27e125efed3b7a995eb8882d"
    },
    "LoggingConfig": {
        "LogFormat": "Text",
        "LogGroup": "/aws/lambda/grc-compliance-monitor"
    }
```

```
GRC208-AWS-Capstone-Project $ aws lambda get-function --function-name grc-compliance-monitor --query "Configuration.{Name:FunctionName,Runtime:Runtime,State:State,Role:Role}" --output table --region us-east-1
-----------------------------------------------------------------------------------------------------
|                                                GetFunction                                          |
+-----------------------+-------------------------------------------------------------+-------------+----------+
|         Name          |                                Role                         |   Runtime   |  State   |
+-----------------------+-------------------------------------------------------------+-------------+----------+
|  grc-compliance-monitor|  arn:aws:iam::768499314195:role/grc-lambda-execution-role  |  python3.11 |  Active  |
+-----------------------+-------------------------------------------------------------+-------------+----------+
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws lambda get-function --function-name grc-compliance-monitor --query "Configuratio
-----------------------------------------------------------------------------------------------------
|                                                GetFunction                                          |
+-----------------------+-------------------------------------------------------------+-------------+----------+
|         Name          |                                Role                         |   Runtime   |  State   |
+-----------------------+-------------------------------------------------------------+-------------+----------+
|  grc-compliance-monitor|  arn:aws:iam::768499314195:role/grc-lambda-execution-role  |  python3.11 |  Active  |
+-----------------------+-------------------------------------------------------------+-------------+----------+
GRC208-AWS-Capstone-Project $
```

# CloudWatch Alarms

```
GRC208-AWS-Capstone-Project $ aws cloudwatch put-metric-alarm --alarm-name "grc-rds-high-cpu" --alarm-description "Alert if RDS CPU exceeds 80%" --metric-name CPUUtilization --namespace AWS/RDS --statistic Average --per
iod 300 --threshold 80 --comparison-operator GreaterThanThreshold --evaluation-periods 2 --dimensions Name=DBInstanceIdentifier,Value=grc-capstone-db --region us-east-1
GRC208-AWS-Capstone-Project $ aws cloudwatch describe-alarms --query "MetricAlarms[].AlarmName" --output text --region us-east-1
```

## CLOUDTRAIL AUDIT

```
GRC208-AWS-Capstone-Project $ # Create trail
GRC208-AWS-Capstone-Project $ aws cloudtrail create-trail --name grc-trail --s3-bucket-name grc-cloudtrail-logs-768499314195 --region us-east-1
{
    "Name": "grc-trail",
    "S3BucketName": "grc-cloudtrail-logs-768499314195",
    "IncludeGlobalServiceEvents": true,
    "IsMultiRegionTrail": false,
    "TrailARN": "arn:aws:cloudtrail:us-east-1:768499314195:trail/grc-trail",
    "LogFileValidationEnabled": false,
    "IsOrganizationTrail": false
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Start logging
GRC208-AWS-Capstone-Project $ aws cloudtrail start-logging --name grc-trail --region us-east-1
```

```
GRC208-AWS-Capstone-Project $ aws cloudtrail describe-trails --query "trailList[0].{Name:Name,Logging:IsLogging}" --output table --region us-east-1
-------------------------
|     DescribeTrails     |
+----------+-------------+
| Logging  |    Name     |
+----------+-------------+
| None     | grc-trail   |
+----------+-------------+
GRC208-AWS-Capstone-Project $
```


```
-------------------------
|     DescribeTrails     |
+----------+-------------+
| Logging  |    Name     |
+----------+-------------+
| None     | grc-trail   |
+----------+-------------+
GRC208-AWS-Capstone-Project $
```


## EVENTBRIDGE

```
GRC208-AWS-Capstone-Project $ # Create rule
GRC208-AWS-Capstone-Project $ aws events put-rule --name grc-compliance-check --schedule-expression "rate(1 hour)" --state ENABLED --region us-east-1
{
    "RuleArn": "arn:aws:events:us-east-1:768499314195:rule/grc-compliance-check"
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Add Lambda target
GRC208-AWS-Capstone-Project $ aws events put-targets --rule grc-compliance-check --targets "Id"="1","Arn"="arn:aws:lambda:us-east-1:768499314195:function:grc-compliance-monitor" --region us-east-1
{
    "FailedEntryCount": 0,
    "FailedEntries": []
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Add Lambda permission
GRC208-AWS-Capstone-Project $ aws lambda add-permission --function-name grc-compliance-monitor --statement-id grc-eventbridge --action lambda:InvokeFunction --principal events.amazonaws.com --source-arn arn:aws:events:u
s-east-1:768499314195:rule/grc-compliance-check --region us-east-1
{
    "Statement": "{\"Sid\":\"grc-eventbridge\",\"Effect\":\"Allow\",\"Principal\":{\"Service\":\"events.amazonaws.com\"},\"Action\":\"lambda:InvokeFunction\",\"Resource\":\"arn:aws:lambda:us-east-1:768499314195:function
:grc-compliance-monitor\",\"Condition\":{\"ArnLike\":{\"AWS:SourceArn\":\"arn:aws:events:us-east-1:768499314195:rule/grc-compliance-check\"}}}"
}
```

```
GRC208-AWS-Capstone-Project $ aws events describe-rule --name grc-compliance-check --query "{Name:Name,State:State,ScheduleExpression:ScheduleExpression}" --output table --region us-east-1
-----------------------------------------------------------
|                       DescribeRule                      |
+-----------------------+----------------------+----------+
|         Name          | ScheduleExpression   |  State   |
+-----------------------+----------------------+----------+
|  grc-compliance-check |  rate(1 hour)        |  ENABLED |
+-----------------------+----------------------+----------+
```

```
GRC208-AWS-Capstone-Project $ aws events describe-rule --name gr
-----------------------------------------------------------
|                       DescribeRule                      |
+-----------------------+----------------------+----------+
|         Name          | ScheduleExpression   |  State   |
+-----------------------+----------------------+----------+
|  grc-compliance-check |  rate(1 hour)        |  ENABLED |
+-----------------------+----------------------+----------+
GRC208-AWS-Capstone-Project $
```

## RDS MANUAL SNAPSHOT

```
GRC208-AWS-Capstone-Project $ aws rds create-db-snapshot --db-instance-identifier grc-capstone-db --db-snapshot-identifier grc-capstone-snapshot-$(date +%Y%m%d) --region us-east-1
{
    "DBSnapshot": {
        "DBSnapshotIdentifier": "grc-capstone-snapshot-20260331",
        "DBInstanceIdentifier": "grc-capstone-db",
        "Engine": "mysql",
        "AllocatedStorage": 20,
        "Status": "creating",
        "Port": 3306,
        "AvailabilityZone": "us-east-1c",
        "VpcId": "vpc-0fee4e290988768fb",
        "InstanceCreateTime": "2026-03-31T19:50:06.606000+00:00",
        "MasterUsername": "grcadmin",
        "EngineVersion": "8.0.44",
        "LicenseModel": "general-public-license",
        "SnapshotType": "manual",
        "StorageThroughput": 0,
        "OptionGroupName": "default:mysql-8-0",
        "PercentProgress": 0,
        "StorageType": "gp2",
        "Encrypted": false,
        "BackupRetentionPeriod": 0,
        "PreferredBackupWindow": "05:05-05:35",
        "DBSnapshotArn": "arn:aws:rds:us-east-1:768499314195:snapshot:grc-capstone-snapshot-20260331",
        "IAMDatabaseAuthenticationEnabled": false,
        "ProcessorFeatures": [],
        "DbiResourceId": "db-CDFCJSPS3AWKO3FECLIBSEZXJA",
        "TagList": [],
        "SnapshotTarget": "region",
        "DedicatedLogVolume": false
    }
}
GRC208-AWS-Capstone-Project $
```


```
"DBSnapshot": {
    "DBSnapshotIdentifier": "grc-capstone-snapshot-20260331",
    "DBInstanceIdentifier": "grc-capstone-db",
    "Engine": "mysql",
    "AllocatedStorage": 20,
    "Status": "creating",
    "Port": 3306,
    "AvailabilityZone": "us-east-1c",
    "VpcId": "vpc-0fee4e290988768fb",
    "InstanceCreateTime": "2026-03-31T19:50:06.606000+00:00",
    "MasterUsername": "grcadmin",
    "EngineVersion": "8.0.44",
    "LicenseModel": "general-public-license",
    "SnapshotType": "manual",
    "StorageThroughput": 0,
    "OptionGroupName": "default:mysql-8-0",
    "PercentProgress": 0,
    "StorageType": "gp2",
    "Encrypted": false,
    "BackupRetentionPeriod": 0,
    "PreferredBackupWindow": "05:05-05:35",
    "DBSnapshotArn": "arn:aws:rds:us-east-1:768499314195:snapshot:grc-capstone-snapshot-20260331",
    "IAMDatabaseAuthenticationEnabled": false,
    "ProcessorFeatures": [],
    "DbiResourceId": "db-CDFCJSPS3AWKO3FECLIBSEZXJA",
    "TagList": [],
    "SnapshotTarget": "region",
    "DedicatedLogVolume": false
}
```


```
GRC208-AWS-Capstone-Project $ aws rds describe-db-snap
-----------------------------------------------
|              DescribeDBSnapshots            |
+---------------------------------+------------+
|               ID                |  Status    |
+---------------------------------+------------+
|  grc-capstone-snapshot-20260331 |  creating  |
+---------------------------------+------------+
GRC208-AWS-Capstone-Project $
```

## .env File Creation

```
GRC208-AWS-Capstone-Project $ cat > .env << 'EOF'
> # GRC208 Capstone - Environment Configuration
> AWS_REGION=us-east-1
> AWS_ACCOUNT_ID=768499314195
> DB_HOST=grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com
> DB_PORT=3306
> DB_NAME=grcdb
> DB_USER=grcadmin
> DB_PASSWORD=MyPassword123
> EVIDENCE_BUCKET=grc-capstone-evidence-bucket-768499314195
> COMPLIANCE_REPORTS_BUCKET=grc-capstone-compliance-reports-768499314195
> LAMBDA_FUNCTION_NAME=grc-compliance-monitor
> LAMBDA_ROLE_ARN=arn:aws:iam::768499314195:role/grc-lambda-execution-role
> NETWORK_STACK=grc-capstone-network-stack
> DATABASE_STACK=grc-capstone-database-stack
> EOF
GRC208-AWS-Capstone-Project $ cat .env
# GRC208 Capstone - Environment Configuration
AWS_REGION=us-east-1
AWS_ACCOUNT_ID=768499314195
DB_HOST=grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com
DB_PORT=3306
DB_NAME=grcdb
DB_USER=grcadmin
DB_PASSWORD=MyPassword123
EVIDENCE_BUCKET=grc-capstone-evidence-bucket-768499314195
COMPLIANCE_REPORTS_BUCKET=grc-capstone-compliance-reports-768499314195
LAMBDA_FUNCTION_NAME=grc-compliance-monitor
LAMBDA_ROLE_ARN=arn:aws:iam::768499314195:role/grc-lambda-execution-role
NETWORK_STACK=grc-capstone-network-stack
DATABASE_STACK=grc-capstone-database-stack
GRC208-AWS-Capstone-Project $
```

## Run Test Suite

```
DATABASE_STACK=grc-capstone-database-stack
GRC208-AWS-Capstone-Project $ python3 test_cases.py
test_compliance_percentage_calculation (__main__.TestComplianceMonitoring)
Test compliance percentage calculation ... ok
test_compliance_report_generation (__main__.TestComplianceMonitoring)
Test compliance report generation ... ok
test_non_compliant_rules_detection (__main__.TestComplianceMonitoring)
Test detection of non-compliant rules ... ok
test_risk_level_classification (__main__.TestRiskAssessment)
Test risk level classification based on compliance ... ok
test_risk_matrix_values (__main__.TestRiskAssessment)
Test risk matrix scoring values ... ok
test_risk_score_calculation (__main__.TestRiskAssessment)
Test risk score calculation ... ok
test_asset_classification_validation (__main__.TestDataValidation)
Test asset classification validation ... ok
test_control_id_format_validation (__main__.TestDataValidation)
Test control ID format validation ... ok
test_criticality_level_validation (__main__.TestDataValidation)
Test criticality level validation ... ok
test_risk_status_validation (__main__.TestDataValidation)
Test risk status validation ... ok
test_compliance_snapshot_structure (__main__.TestDatabaseOperations)
Test compliance snapshot data structure ... ok
test_control_record_structure (__main__.TestDatabaseOperations)
Test control record structure ... ok
test_risk_register_structure (__main__.TestDatabaseOperations)
Test risk register data structure ... ok
test_framework_control_count (__main__.TestComplianceFrameworks)
Test framework control count ... ok
test_framework_version_format (__main__.TestComplianceFrameworks)
Test framework version format ... ok
test_audit_log_action_types (__main__.TestAuditLogging)
Test audit log action types ... ok
test_audit_log_entity_types (__main__.TestAuditLogging)
Test audit log entity types ... ok
test_audit_log_entry_structure (__main__.TestAuditLogging)
Test audit log entry structure ... ok
```

```
test_compliance_report_content (__main__.TestReportGeneration)
Test compliance report contains required content ... ok
test_risk_report_content (__main__.TestReportGeneration)
Test risk report contains required content ... ok
test_compliance_to_risk_workflow (__main__.TestIntegration)
Test workflow from compliance check to risk assessment ... ok
test_control_to_audit_workflow (__main__.TestIntegration)
Test workflow from control implementation to audit logging ... ok

----------------------------------------------------------------------
Ran 22 tests in 0.028s

OK
GRC208-AWS-Capstone-Project $
```

## Database Connectivity

```
GRC208-AWS-Capstone-Project $ export DB_ENDPOINT=$(aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].Endpoint.Address" --output text --region us-east-1)
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Verify the endpoint
GRC208-AWS-Capstone-Project $ echo "RDS Endpoint: $DB_ENDPOINT"
RDS Endpoint: grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com
GRC208-AWS-Capstone-Project $
```


```
GRC208-AWS-Capstone-Project $ export DB_ENDPOINT=$(aws rds describe-db-instances
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Verify the endpoint
GRC208-AWS-Capstone-Project $ echo "RDS Endpoint: $DB_ENDPOINT"
RDS Endpoint: grc-capstone-db.cc5w8ia2urid.us-east-1.rds.amazonaws.com
GRC208-AWS-Capstone-Project $
```


## Database Connection

## LOAD SAMPLE DATA

```
GRC208-AWS-Capstone-Project $ # Get VPC and subnet info from your RDS instance
GRC208-AWS-Capstone-Project $ VPC_ID=$(aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].DBSubnetGroup.VpcId" --output text --region us-east-1)
GRC208-AWS-Capstone-Project $ SUBNETS=$(aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].DBSubnetGroup.Subnets[*].SubnetIdentifier" --output text --region us-east-1 | tr '
 ',')
GRC208-AWS-Capstone-Project $ RDS_SG=$(aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].VpcSecurityGroups[0].VpcSecurityGroupId" --output text --region us-east-1)
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ echo "✅ VPC: $VPC_ID"
✅ VPC: vpc-0fee4e290988768fb
GRC208-AWS-Capstone-Project $ echo "✅ Subnets: $SUBNETS"
✅ Subnets: subnet-0a5bca33f33a9a7fa    subnet-0dcd6fc7c0af92213        subnet-0b4e8cfea11c60aaa        subnet-0af36d5f3fbb81c33        subnet-0439e82b9a3be7479        subnet-0aa9778c56406c94f
GRC208-AWS-Capstone-Project $ echo "✅ Security Group: $RDS_SG"
✅ Security Group: sg-011392aa196bd5aab
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ echo "✅ Data loader package created!"
✅ Data loader package created!
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws lambda delete-function --function-name grc-load-sample-data --region us-east-1
{
    "StatusCode": 204
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Clean up local files
GRC208-AWS-Capstone-Project $ rm -rf data-loader data-loader.zip
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ echo ""

GRC208-AWS-Capstone-Project $ echo "✅ Data loading complete!"
✅ Data loading complete!
GRC208-AWS-Capstone-Project $
```

## COMPLIANCE MONITORING
## AWS CONFIG

```
GRC208-AWS-Capstone-Project $ # Verify the bucket policy
GRC208-AWS-Capstone-Project $ aws s3api get-bucket-policy --bucket grc-config-bucket-768499314195 --region us-east-1 --query "Policy" --output text | python3 -m json.tool
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AWSConfigBucketPermissionsCheck",
            "Effect": "Allow",
            "Principal": {
                "Service": "config.amazonaws.com"
            },
            "Action": "s3:GetBucketAcl",
            "Resource": "arn:aws:s3:::grc-config-bucket-768499314195"
        },
        {
            "Sid": "AWSConfigBucketExistenceCheck",
            "Effect": "Allow",
            "Principal": {
                "Service": "config.amazonaws.com"
            },
            "Action": "s3:ListBucket",
            "Resource": "arn:aws:s3:::grc-config-bucket-768499314195"
        },
        {
            "Sid": "AWSConfigBucketDelivery",
            "Effect": "Allow",
            "Principal": {
                "Service": "config.amazonaws.com"
            },
            "Action": "s3:PutObject",
            "Resource": "arn:aws:s3:::grc-config-bucket-768499314195/config/*"
        }
    ]
}
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws configservice describe-delivery-channels --region us-east-1
{
    "DeliveryChannels": [
        {
            "name": "default",
            "s3BucketName": "grc-config-bucket-768499314195",
            "s3KeyPrefix": "config"
        }
    ]
}
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ aws configservice describe-configuration-recorder-status --region us-east-1
{
    "ConfigurationRecordersStatus": [
        {
            "arn": "arn:aws:config:us-east-1:768499314195:configuration-recorder/default/fbca5rj92vdneqj8",
            "name": "default",
            "lastStartTime": "2026-03-31T22:07:42.841000+00:00",
            "recording": true,
            "lastStatus": "SUCCESS",
            "lastStatusChangeTime": "2026-03-31T22:07:53.337000+00:00"
        }
    ]
}
GRC208-AWS-Capstone-Project $
```

```
GRC208-AWS-Capstone-Project $ # Wait 2 minutes for Config to discover resources
GRC208-AWS-Capstone-Project $ echo "⏳ Waiting for AWS Config to discover resources (2 minutes)..."
⏳ Waiting for AWS Config to discover resources (2 minutes)...
GRC208-AWS-Capstone-Project $ sleep 120


GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check discovered resource counts
GRC208-AWS-Capstone-Project $ aws configservice get-discovered-resource-counts --region us-east-1
{
    "totalDiscoveredResources": 0,
    "resourceCounts": []
}
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $
```

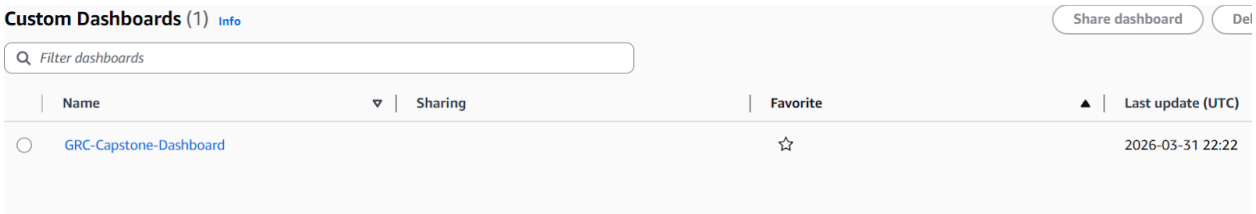
```
GRC208-AWS-Capstone-Project $ aws configservice describe-configuration-recorder-status --region us-east-1 --query "ConfigurationRecordersStatus[0].{Name:name,Recording:recording,LastStatus:lastStatus}" --output table
-----------------------------------------
|  DescribeConfigurationRecorderStatus  |
+------------+-----------+-------------+
| LastStatus |   Name    |  Recording  |
+------------+-----------+-------------+
|  SUCCESS   |  default  |  True       |
+------------+-----------+-------------+
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check delivery channel exists
GRC208-AWS-Capstone-Project $ aws configservice describe-delivery-channels --region us-east-1 --query "DeliveryChannels[0].{Name:name,Bucket:s3BucketName}" --output table
-------------------------------------------------
|           DescribeDeliveryChannels            |
+-----------------------------------+----------+
|              Bucket               |  Name    |
+-----------------------------------+----------+
|  grc-config-bucket-768499314195   |  default |
+-----------------------------------+----------+
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check role is attached correctly
GRC208-AWS-Capstone-Project $ aws configservice describe-configuration-recorders --region us-east-1 --query "ConfigurationRecorders[0].{Name:name,Role:roleARN}" --output table
```

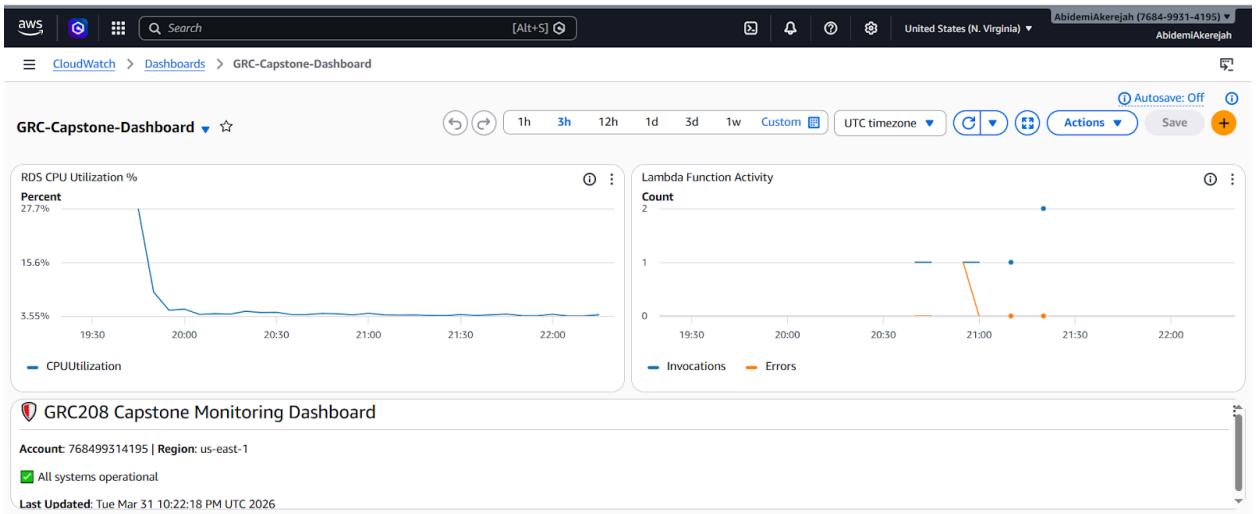
```
GRC208-AWS-Capstone-Project $ aws configservice describe-configur
-----------------------------------------
|  DescribeConfigurationRecorderStatus  |
+------------+-----------+-------------+
| LastStatus |   Name    |  Recording  |
+------------+-----------+-------------+
|  SUCCESS   |  default  |  True       |
+------------+-----------+-------------+
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check delivery channel exists
GRC208-AWS-Capstone-Project $ aws configservice describe-delivery
-------------------------------------------------
|           DescribeDeliveryChannels            |
+-----------------------------------+----------+
|              Bucket               |  Name    |
+-----------------------------------+----------+
|  grc-config-bucket-768499314195   |  default |
+-----------------------------------+----------+
GRC208-AWS-Capstone-Project $
GRC208-AWS-Capstone-Project $ # Check role is attached correctly
GRC208-AWS-Capstone-Project $ aws configservice describe-configur
```

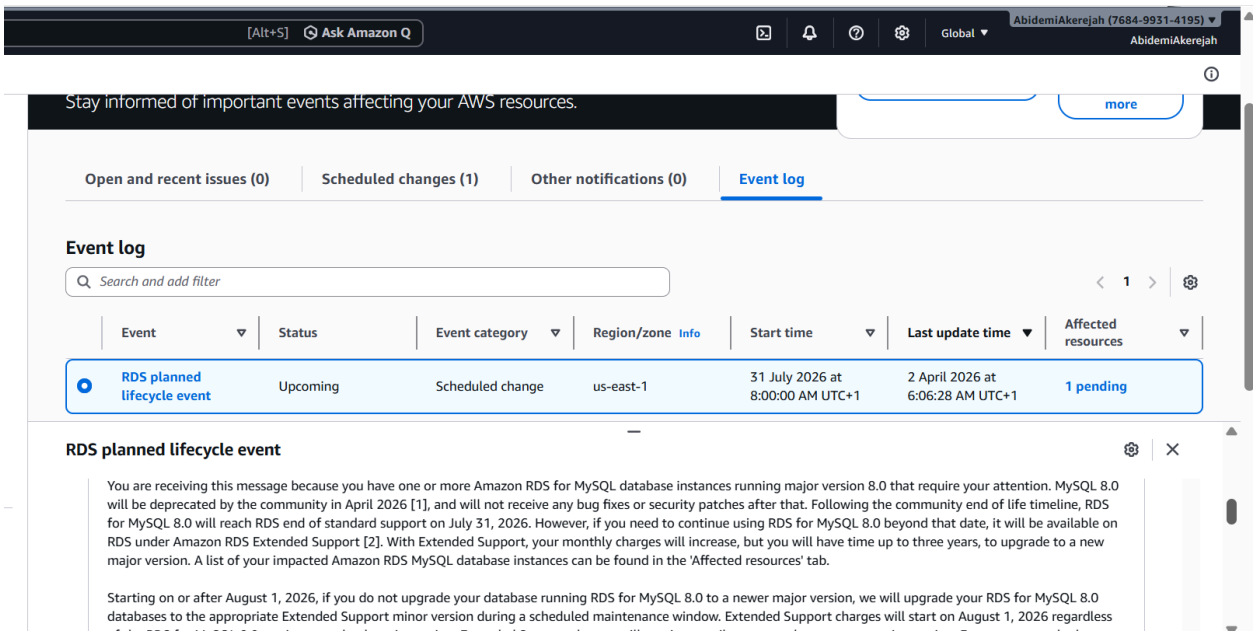
```
GRC208-AWS-Capstone-Project $ aws configservice describe-configuration-recorders --
-------------------------------------------------------------
|                DescribeConfigurationRecorders               |
+---------+-----------------------------------------------------+
|  Name   |                        Role                         |
+---------+-----------------------------------------------------+
| default|  arn:aws:iam::768499314195:role/grc-config-role      |
+---------+-----------------------------------------------------+
```

## CLOUDWATCH DASHBOARD

```
>         "properties": {
>           "markdown": "## 🛡️ GRC208 Capstone Monitoring Dashboard\n\n**Account**: 768499314195 | **Region**: us-east-1\n\n✅ All systems operational\n\n**Last Updated**: '"$(date)"'"
>         }
>       }
>     ]
>   }' \
>   --region us-east-1
{
    "DashboardValidationMessages": []
}
GRC208-AWS-Capstone-Project $ # Verify dashboard exists
GRC208-AWS-Capstone-Project $ aws cloudwatch get-dashboard --dashboard-name GRC-Capstone-Dashboard --region us-east-1 --query "DashboardName" --output text
GRC-Capstone-Dashboard
GRC208-AWS-Capstone-Project $
```




## Personal AWS Health Event -RDS MySQL 8.0

## evidence/rds-lifecycle-notification-screenshot